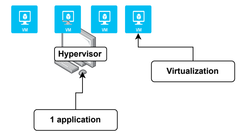


Class 8:Traditional IT Vs Cloud

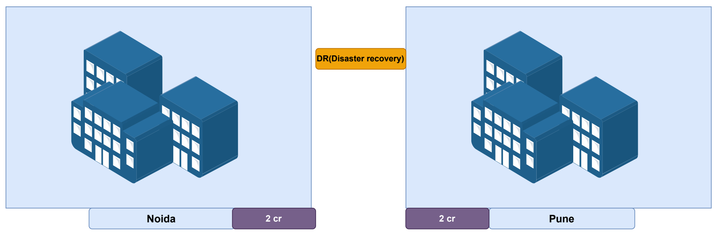
Traditional IT

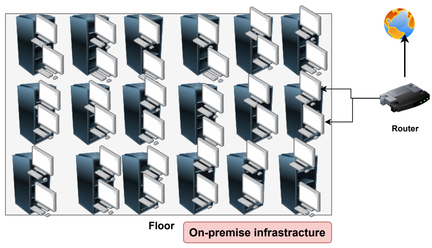
Data Center

Application jabong

Middleware

Infrastructure





On-premise infrastructure

Traditional IT / Data Center ki Problems

- **High Cost (Bahut mehenga)**
  - Server, storage, networking sab khud kharidna padta tha
  - Maintenance + power + cooling ka extra cost
- **Scalability Issue**
  - Demand badhne par turant server add nahi kar sakte
  - New hardware purchase me time lagta tha
- **Downtime Risk**
  - Hardware fail ho gaya toh application down
  - High availability maintain karna difficult
- **Manual Work (Automation nahi)**
  - Deployment, patching, monitoring sab manual
  - Human error ke chances zyada
- **Disaster Recovery Problem**
  - Backup aur DR setup costly aur complex
  - Data loss ka risk high

Problems

1-High Cost(Capital+Operation)

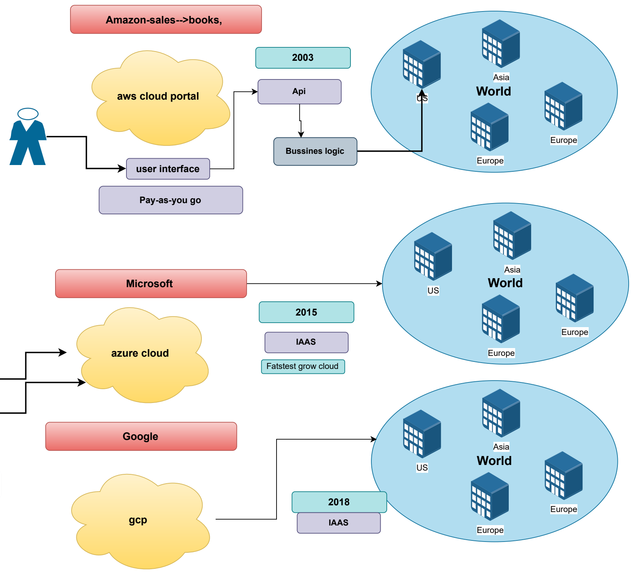
1-Intial Setup bahut coslty hota hai(Servers,Cooling,Networking)-Copex
2-Electricity+Maintaenance ka cost continue hota hai
3-Hardware Old hone par fir change karonge

2-Power & Cooling issue

1-Continue power supply chahiye(UPS+Generator)
2-Cooling ke lie AC Chahiye-servers heat generate karte hai
3-Energy consumtion bahut high clear hota

3-Mantenance Problem

- **Space & Infrastructure Dependency**
  - Physical data center ke liye jagah chahiye
  - Cooling, racks, cabling sab manage karna padta
- **Security Challenges**
  - Physical + network security dono manage karna
  - Dedicated security setup required
- **Slow Deployment**
  - Server setup me days ya weeks lag jate the
  - Business agility slow
- **Resource Utilization Poor**
  - Server idle rehte the (under-utilization)
  - Cost waste hota tha
- **Maintenance Headache**
  - Hardware failure, updates, replacements
  - Dedicated IT team required



Infrastructure

Client 20%

service base company 20%

TCS,hcl,infosys,TechM

Oracle cloud

IBM Cloud

Alibaba Cloud

Alibaba Cloud

Virtualization

Hypervisor Tool

VMware-vmware workstation,esxi

Micorsoft-Hyper-v

Application

Middleware

Infrastructure

Deployment Model Of Cloud Computing

Public Cloud

Owned & Managed by Cloud Provider (aws,gcp,azure)

Shared Infrastructure(default) (Multi-tenant Mostly)

Logical Isolate

Private Cloud

Owned & Used by single organization

Hybrid Cloud